

SHAH NAWAAS

Software Engineer | Data Scientist | AI/ML Engineer | Full Stack Developer
+91 8778487141 · shahnu2109@gmail.com · github.com/shahnawaas · linkedin.com/in/shah-nawaas

PROFESSIONAL SUMMARY

B.Tech graduate in Artificial Intelligence & Data Science with hands-on experience in software engineering, machine learning, backend API development, and full-stack web development. Skilled in Python, Java, FastAPI, Spring Boot, TensorFlow, PyTorch, React.js, SQL, and REST APIs. Experienced in building end-to-end ML pipelines, developing scalable backend services, and creating production-oriented web applications through academic and personal projects. Passionate about solving real-world problems with technology and seeking a full-time Software Engineer, Backend Developer, Full-Stack Developer, AI/ML Engineer, or Data Analyst role where I can contribute, learn, and grow.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Frontend: React.js, Angular, HTML5, CSS3, Responsive Web Design

Backend: FastAPI, Spring Boot, Node.js, REST APIs, SQLAlchemy, Pydantic

Databases: PostgreSQL, MongoDB, MySQL

AI / Machine Learning: TensorFlow, Keras, PyTorch, Scikit-Learn, Pandas, NumPy, OpenCV, NLP, Computer Vision, Deep Learning, Anomaly Detection, Isolation Forest, Autoencoders

Cloud & Deployment: Render, Neon PostgreSQL, Git, GitHub Actions

Tools: Git, GitHub, Postman, Swagger, Streamlit, Power BI, Jupyter Notebook, Google Colab, VS Code

CODING & PROFESSIONAL PROFILES

GitHub: github.com/shahnawaas | **LinkedIn:** linkedin.com/in/shah-nawaas

EDUCATION

B.Tech — Artificial Intelligence & Data Science

Sep 2022 – May 2026

Mohamed Sathak AJ College of Engineering, Chennai

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Machine Learning, Artificial Intelligence

PROJECTS

AI-Based Anomaly Detection System | [Python](#) · [Scikit-Learn](#) · [TensorFlow](#) · [Keras](#) · [Streamlit](#)

- **Problem:** Traditional signature-based intrusion detection systems fail to identify novel zero-day attacks due to reliance on known threat patterns.
- **Solution:** Architected an unsupervised ML pipeline combining Isolation Forest and Autoencoder models on the KDD Cup 1999 dataset to detect anomalous network traffic without labeled attack data.
- **Engineered** a multi-stage preprocessing pipeline — feature encoding, normalization, and noise reduction across 41 network attributes — improving model accuracy and convergence speed.
- **Impact:** Evaluated using Precision, Recall, F1-Score, and Accuracy metrics; deployed a real-time interactive Streamlit web application enabling live anomaly prediction on uploaded network data.

Inventory Order Management System | [FastAPI](#) · [PostgreSQL](#) · [SQLAlchemy](#) · [Pydantic](#) · [Render](#) · [Neon](#)

- **Problem:** E-commerce systems risk overselling limited stock when multiple users submit simultaneous purchase requests, leading to data inconsistency and poor user experience.
- **Solution:** Developed a production-grade RESTful backend using FastAPI and PostgreSQL implementing row-level locking (SELECT FOR UPDATE / with_for_update()) to serialize concurrent order transactions.
- **Implemented** idempotency key handling to prevent duplicate order creation on retried API requests; enforced structured request validation using Pydantic schemas throughout the API layer.
- **Impact:** Achieved zero oversell incidents under concurrent load testing; deployed live on Render with Neon cloud PostgreSQL; validated via Swagger documentation and Postman test suites.

Shopping Mall Management System | [Spring Boot](#) · [Angular](#) · [PostgreSQL](#) · [REST APIs](#) · [Postman](#)

- **Problem:** Retail management platforms require reliable, performant APIs to synchronize customer data, store inventory, and cross-team workflows without integration errors.
- **Solution:** Designed and implemented 15+ RESTful API endpoints using Spring Boot for customer and store data management; integrated with an Angular frontend ensuring consistent data flow and sub-200ms response times.

- **Conducted** end-to-end QA testing using Postman, identifying and resolving 10+ critical API defects; defined an API contract that eliminated cross-team integration issues during development.
- **Impact:** Delivered a fully tested, contract-driven full-stack application with validated backend performance and zero post-integration API regressions.

Responsive Restaurant Web Application | [React.js](#) · [JavaScript](#) · [HTML5](#) · [CSS3](#) · [CSS Grid](#)

- **Problem:** Restaurant websites typically suffer from slow load times, broken layouts on mobile, and Safari/iOS rendering bugs that degrade user experience across devices.
- **Solution:** Built a multi-module React.js application implementing lazy loading, CSS Grid auto-fill layouts, and -webkit- CSS prefix fallbacks to ensure cross-browser compatibility.
- **Resolved** Safari/iOS-specific rendering inconsistencies and conducted structured QA across Chrome, Firefox, Safari, and mobile viewports to validate pixel-accurate responsiveness.
- **Impact:** Achieved measurably improved page load performance via lazy loading and pixel-accurate rendering across all major browsers and screen sizes.

WORK EXPERIENCE

Full Stack Developer Intern | *Unistrix IT Solutions*

Jan 2026 – May 2026

- **Developed** and maintained full-stack web applications using React.js, FastAPI, Spring Boot, and SQL, delivering features across both frontend and backend layers.
- **Built** and integrated REST APIs, assisted with relational database design, and implemented frontend and backend features following coding standards and Git-based version control workflows.
- **Performed** debugging, testing, and deployment tasks; collaborated with senior developers on real-world projects, gaining end-to-end SDLC exposure across planning, development, QA, and release stages.

Web Development & Digital Marketing Intern | *Evoulth Digital Pvt Ltd*

Jan 2023 – Jun 2023

- **Delivered** 5+ end-to-end client web projects, applying QA validation and adapting to evolving requirements, consistently meeting project delivery timelines.
- **Collaborated** on data-driven social media campaigns leveraging Google Analytics and the Google product suite, contributing to measurable improvements in online brand engagement and reach.

CERTIFICATIONS & AWARDS

Java Full Stack — TNS Foundation | **MERN Full Stack** — Revamp Academy

IBM Career Education Program — CognitiveClass | **Power BI** — IBM

Hackathon Coordinator — Star Systems (24-hour event, 100+ participants)